

Large-scale dynamical model approximation and its applications

... a linear time invariant framework

Charles Poussot-Vassal

July, 12th, 2019
HDR defense



A.C. Antoulas	Rice University	Houston, Texas, USA	Reviewer
P. Apkarian	Onera	Toulouse, France	
G. Garcia	INSA - LAAS	Toulouse, France	
S. Grundel	Max Planck Institute	Magdeburg, Germany	
M. Olivi	INRIA Sophia-Antipolis	Nice, France	
C. Prieur	CNRS Gipsa-lab	Grenoble, France	Reviewer
K. Willcox	University of Texas	Austin, Texas, USA	Reviewer

Outline

1. Short resume and activities at **Onera**

⇒ research environment and context

2. Framework introduction on dynamical model approximation

⇒ general motivations, notations and some driving examples

3. Interpolation-based dynamical model approximation

⇒ core contributions and applications (delay and limited frequency range)

4. Some on-going researches and future targets

⇒ try to connect some recent work and ideas

5. Conclusions

⇒ what to keep in mind...

Outline

Resume

Path toward "dynamical model reduction and approximation"

Research and projects environment at Onera

Dynamical models and approximation

Dynamical model approximation and interpolation

Some open research paths

Conclusions

Resume

Path toward "dynamical model reduction and approximation"

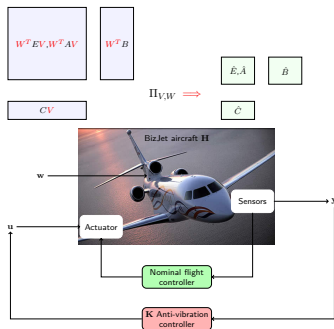
- 2005 Eng. degree, **ESISAR, Grenoble INP**
M.Sc. degree **Lund University**
- 2008 Ph.D. degree, **Grenoble INP**
- 2009 Post-doctoral, **Politecnico di Milano**



Resume

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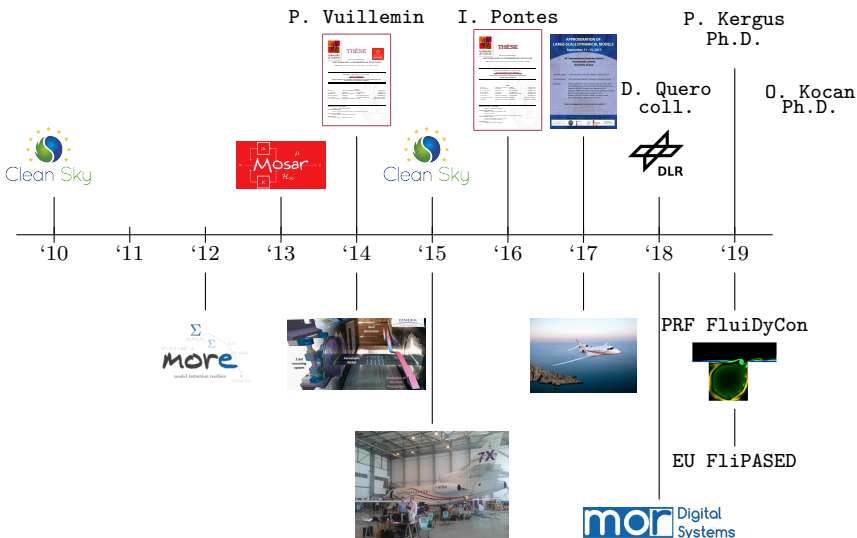


- 2009 Researcher **Onera** DSCD, CDIN
→ Control and integration team
- 2017 Researcher **Onera** DTIS, COVNI
→ Control, verification, num. integr.
- 10 years at **Onera** Toulouse working on **civil-ian aircraft** and **model approximation**

Teaching activities at INSA Toulouse and ISAE on **control / modelling labs**

Resume

Research and projects environment at Onera



Outline

Resume

Dynamical models and approximation

- Model approximation, what for?

- Considered class of linear dynamical models and I/O data

- Talk's objectives and messages

- Illustrated with 3 driving examples

Dynamical model approximation and interpolation

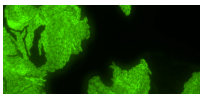
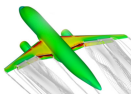
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Dynamical models and approximation

Model approximation, what for?

Digitalisation and computer-based modelling and studies are crucial steps for understanding, simulating, controlling system, concept or physical phenomena



Problem: numerical dynamical models are too complex and parameter dependent

Finite machine precision, computational burden and memory management

- ▶ induces important time consumption
- ▶ generate inaccurate results

And actual numerical tools

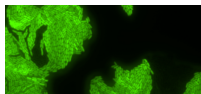
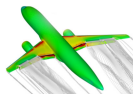
- ▶ limit the use of class and complexity models

Solution: provide robust and efficient mathematical and numerical tools to simplify (linear) dynamical models

Dynamical models and approximation

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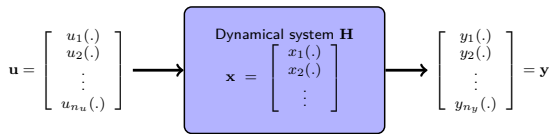
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Dynamical models and approximation

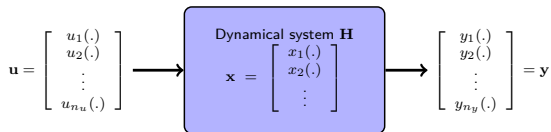
Considered class of linear dynamical models and I/O data



$$\begin{aligned} \mathbf{u}(s) \quad \mathbf{H}(s) &= \mathbf{y}(s) \\ \mathbf{u}(i\omega_i) \quad \Phi_i &= \mathbf{y}(i\omega_i) \end{aligned}$$

Dynamical models and approximation

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Finite dynamical models have a **finite** number of singularities, e.g.

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Infinite dynamical models have an **infinite** number of singularities, e.g.

$$\mathbf{H}(s) = \frac{1}{s + e^{-s}} \in \mathcal{L}_2$$

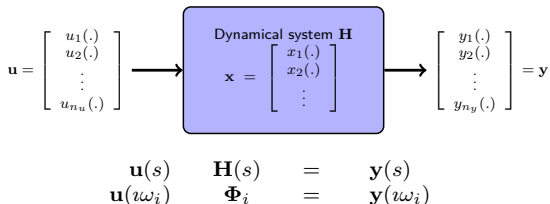
Input/output data have **no structure**, $i = 1, \dots, N$ e.g.

$$\Phi_i = \frac{\mathbf{y}(\omega_i)}{\mathbf{u}(\omega_i)} \in \mathbb{C}^{n_y \times n_u}$$

$$\mathcal{L}_p : \{ \mathbf{H} : \mathbb{C} \rightarrow \mathbb{C}^{n_y \times n_u}, \|\mathbf{H}\|_p < \infty \} \text{ and } \mathcal{RL}_p : \{ \text{rational } \mathcal{L}_p \text{ functions} \}$$

Dynamical models and approximation

Considered class of linear dynamical models and I/O data



Model approximation paradigm seeks for an **approximation** $\hat{\mathbf{H}}$ which:

- ▶ is **uniformly "close"**, i.e. for all $\mathbf{u}$,

$$(\mathbf{H} - \hat{\mathbf{H}})\mathbf{u} \text{ or } (\Phi_i - \hat{\mathbf{H}}(j\omega_i))\mathbf{u}$$

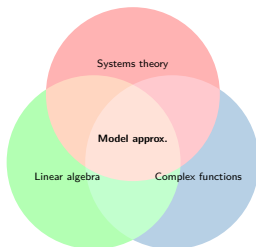
is "small" in an appropriate sense,

- ▶ preserves some **properties**, e.g. stability, passivity etc.
- ▶ model is obtained with a **numerically robust and stable** procedure.

The interpolation framework is a (good) solution...

Dynamical models and approximation

Talk's objectives and messages



Interpolation-based framework is suited for

- ▶ for model-based reduction and approximation
- ▶ for data-driven model construction

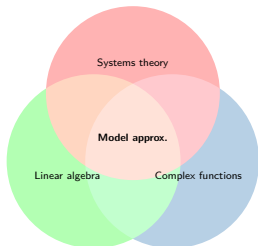
and **may be** a pivotal tool for **complex functions**

- ▶ ... stability estimation
- ▶ ... performance estimation
- ▶ ... control design

⇒ **Central tool for dynamical systems analysis**

Dynamical models and approximation

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Focus on interpolation-based

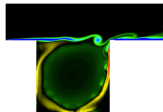
- ▶ approximation by a **I/O delay structured** model [Ph.D. I. Pontes]
- ▶ approximation over **frequency-limited** range [Ph.D. P. Vuillemin]
- ▶ **control design** applications [Ph.D. P. Kergus]
- ▶ **stability and margin estimation**

Dynamical models and approximation

Illustrated with 3 driving examples

#1 open cavity geometry (flow simulation of a transport phenomena)

- ▶ Very-large scale model $\mathbf{H}_{\#1}(s) = C(sE - A)^{-1}B$
- ▶ Illustrates **I/O delayed model-based** approximation



D. Sipp's open cavity flow.

#2 generic business jet aircraft (vibration control)

- ▶ $\mathbf{H}_{\#2}(s) = C(sI - A)^{-1}B$ and $\{\omega_i, \Phi_i\}_{i=1}^N$
- ▶ Illustrates **frequency-limited** & **data-driven** model approximation



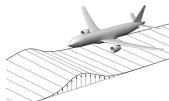
Dassault-Aviation, Falcon 7X.

#3 generic business jet aircraft (gust load control)

- ▶ Multi-delays transfer function $\mathbf{H}_{\#3}$

$$\mathbf{H}(s, \tau_1, \tau_2) = C \left(sE - A_0(\tau_g) - A_1(\tau_g)e^{-\tau_1 s} - A_2(\tau_g)e^{-\tau_2 s} \right)^{-1} B$$

- ▶ Illustrates **infinite dimensional** approximation & **stability estimation**



P. Lancelot et al.,
"Investigation of the
unsteady flow over a
wing gust excitation",
IFASD 2017.

Outline

Resume

Dynamical models and approximation

Dynamical model approximation and interpolation

- Problem formulation

- $\mathcal{H}_2$ model approximation

- $\mathcal{H}_2$ model approximation with I/O delay structure

- $\mathcal{H}_{2,\Omega}$ model approximation

Some open research paths

Conclusions

Dynamical model approximation and interpolation

Problem formulation

Let us consider $\mathbf{H}$, a n_u inputs, n_y outputs linear dynamical system described by the **complex-valued function from $\mathbf{u}$ to $\mathbf{y}$, of order n** (n large or ∞)

$$\mathbf{H} : \mathbb{C} \rightarrow \mathbb{C}^{n_y \times n_u},$$

the model approximation problem consists in finding $\hat{\mathbf{H}}$ of order $r \ll n$

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that **well reproduces the input-output behaviour**

Dynamical model approximation and interpolation

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$$\hat{\mathbf{H}} : \mathbb{C} \rightarrow \mathbb{C}^{n_y \times n_u},$$

that **well reproduces the input-output behaviour** and equipped with realization¹

$$\hat{\mathbf{H}} = \hat{C} (\hat{E}s - \hat{A})^{-1} \hat{B} \quad \text{or} \quad \hat{\mathbf{H}}_d = \hat{\Delta}_o(s) \underbrace{\hat{C} (\hat{E}s - \hat{A})^{-1} \hat{B}}_{\hat{\mathbf{H}}} \hat{\Delta}_i(s)$$

¹ $\hat{\Delta}_i(s)$, $\hat{\Delta}_o(s)$ are the input and output fixed delay operators.

Dynamical model approximation and interpolation

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"Well reproduce..."?

$\hat{\mathbf{H}}$ is a "good" approximation of $\mathbf{H}$ if
for the any driving $\mathbf{u}(t)$, $\mathcal{E}(t) = (\mathbf{H} - \hat{\mathbf{H}})\mathbf{u}(t) = \mathbf{y}(t) - \hat{\mathbf{y}}(t)$ is "small"

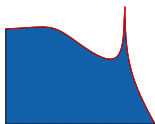
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Dynamical model approximation and interpolation

Problem formulation ($\mathcal{H}_2$ approximation)

$\mathcal{H}_2$ model approximation

$$\hat{\mathbf{H}} := \arg \min_{\substack{\mathbf{G} \in \mathcal{H}_2 \\ \text{rank}(\mathbf{G}) = r \ll n}} \|\mathbf{H} - \mathbf{G}\|_{\mathcal{H}_2}$$



$\mathcal{H}_2$ -norm of $\mathbf{H} - \hat{\mathbf{H}}$

Frequency-domain interpretation:

$$\|\mathbf{H}\|_{\mathcal{H}_2}^2 := \frac{1}{2\pi} \int_{-\infty}^{\infty} \text{tr}(\overline{\mathbf{H}(i\nu)} \mathbf{H}^T(i\nu)) d\nu$$

Time-domain interpretation:

$$\|\mathbf{y}(t) - \hat{\mathbf{y}}(t)\|_{L_\infty} \leq \|\mathbf{H} - \hat{\mathbf{H}}\|_{\mathcal{H}_2} \|\mathbf{u}(t)\|_{L_2}$$



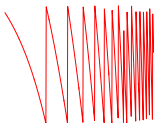
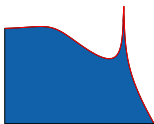
S. Gugercin and A. C. Antoulas and C. A. Beattie, " $\mathcal{H}_2$ Model Reduction for Large Scale Linear Dynamical Systems", SIAM Journal on Matrix Analysis and Applications, 2008, Vol. 30(2), pp. 609-638.

Dynamical model approximation and interpolation

Problem formulation (I/O delays structured $\mathcal{H}_2$ approximation)

Input / output delays structured $\mathcal{H}_2$ model approximation

$$\hat{\mathbf{H}}_d := \arg \min_{\substack{\mathbf{G} \in \mathcal{H}_\infty \\ \text{rank}(\mathbf{G}) = r \ll n}} \|\mathbf{H} - \underbrace{\mathbf{G}}_{\hat{\Delta}_o \hat{\mathbf{H}} \hat{\Delta}_i}\|_{\mathcal{H}_2}$$



$\mathcal{H}_2$ -norm of $\mathbf{H} - \hat{\mathbf{H}}e^{-\hat{\tau}s}$

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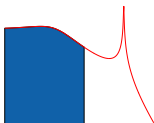
I. Pontes Duff, C. P-V. and C. Seren, " *$\mathcal{H}_2$ -optimal model approximation by input / output-delay structured reduced order models*", Systems & Control Letters, 2018, Vol. 117, pp. 60-67.

Dynamical model approximation and interpolation

Problem formulation ($\mathcal{H}_{2,\Omega}$ approximation)

$\mathcal{H}_{2,\Omega}$ model approximation

$$\hat{\mathbf{H}} := \arg \min_{\substack{\mathbf{G} \in \mathcal{H}_\infty \\ \text{rank}(\mathbf{G}) = r \ll n}} \|\mathbf{H} - \mathbf{G}\|_{\mathcal{H}_{2,\Omega}}$$



$\mathcal{H}_{2,\Omega}$ -norm of $\mathbf{H} - \hat{\mathbf{H}}$

(finite support) Frequency-domain interpretation:

$$\|\mathbf{H}\|_{\mathcal{H}_{2,\Omega}}^2 := \frac{1}{\pi} \int_{\Omega} \text{tr}(\overline{\mathbf{H}(v)} \mathbf{H}^T(v)) dv$$



P. Vuillemin, C. P-V. and D. Alazard, " $\mathcal{H}_2$ optimal and frequency-limited approximation methods for large-scale LTI dynamical systems", in Proc. of the IFAC SSSC, Grenoble, France, 2013, pp. 719-724.

Dynamical model approximation and interpolation

$\mathcal{H}_2$ model approximation ($\hat{\mathbf{H}} := \arg \min ||\mathbf{H} - \mathbf{G}||_{\mathcal{H}_2}$)

If $\hat{\mathbf{H}}$ is solution of the $\mathcal{H}_2$ approximation problem, then

$$\begin{aligned}\mathbf{H}(-\hat{\lambda}_l) &= \hat{\mathbf{H}}(-\hat{\lambda}_l) \\ \mathbf{H}'(-\hat{\lambda}_l) &= \hat{\mathbf{H}}'(-\hat{\lambda}_l)\end{aligned}$$

where $\hat{\lambda}_l := \text{eig}(\hat{E}, \hat{A})^a$ and

$$\mathbf{H}(s) = \sum_{j=1}^n \frac{\phi_j}{s - \lambda_j}$$

$$^a l = 1, \dots, r$$

By considering

$$\begin{aligned}\hat{\mathbf{H}}(s) &= \sum_{k=1}^r \frac{\hat{\phi}_k}{s - \hat{\lambda}_k} \\ &= \hat{\mathbf{c}} (s\hat{E} - \hat{A})^{-1} \hat{\mathbf{b}}\end{aligned}$$

Stands as the central result $\Rightarrow$ but how to determine $\hat{\lambda}_l$?



S. Gugercin and A. C. Antoulas and C. A. Beattie, " *$\mathcal{H}_2$ Model Reduction for Large Scale Linear Dynamical Systems*", SIAM Journal on Matrix Analysis and Applications, 2008, Vol. 30(2), 2008, pp. 609-638.



K. A. Gallivan, A. Vanderoppe, and P. Van-Dooren, "*Model reduction of MIMO systems via tangential interpolation*", SIAM Journal of Matrix Analysis and Application, 2004, Vol. 26(2), pp. 328-349.

Dynamical model approximation and interpolation

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$$\mathbf{H}(s) = \mathbf{c}(sE - A)^{-1} \mathbf{b}$$

↓ **Petrov-Galerkin (interp.)**

$$\hat{\mathbf{H}}(s) = \mathbf{c}V \left(sW^T E V - W^T A V \right)^{-1} W^T \mathbf{b}$$

For some $\sigma_l \in \mathbb{C}$ and $W^T V = I_r$

$$\begin{aligned}\text{span}(V) &\subseteq \left[(\sigma_1 E - A)^{-1} \mathbf{b}, \dots, (\sigma_r E - A)^{-1} \mathbf{b} \right] \\ \text{span}(W) &\subseteq \left[(\sigma_1 E - A)^{-T} \mathbf{c}^T, \dots, (\sigma_r E - A)^{-T} \mathbf{c}^T \right]\end{aligned}$$



S. Gugercin and A. C. Antoulas and C. A. Beattie, " *$\mathcal{H}_2$ Model Reduction for Large Scale Linear Dynamical Systems*", SIAM Journal on Matrix Analysis and Applications, 2008, Vol. 30(2), pp. 609-638.

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↓ **IRKA or TF-IRKA**

$\hat{\mathbf{H}}$ solves the $\mathcal{H}_2$ model approx. pb.

For $\sigma_l \leftarrow -\hat{\lambda}_l$ and $W^T V = I_r$

$$\begin{aligned}\text{span}(V) &\subseteq \left[(-\hat{\lambda}_1 E - A)^{-1} \mathbf{b}, \dots, (-\hat{\lambda}_r E - A)^{-1} \mathbf{b} \right] \\ \text{span}(W) &\subseteq \left[(-\hat{\lambda}_1 E - A)^{-T} \mathbf{c}^T, \dots, (-\hat{\lambda}_r E - A)^{-T} \mathbf{c}^T \right]\end{aligned}$$



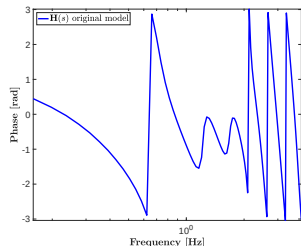
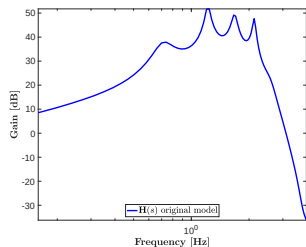
S. Gugercin and A. C. Antoulas and C. A. Beattie, " *$\mathcal{H}_2$ Model Reduction for Large Scale Linear Dynamical Systems*", SIAM Journal on Matrix Analysis and Applications, 2008, Vol. 30(2), pp. 609-638.

Dynamical model approximation and interpolation

$\mathcal{H}_2$ model approximation ($H_{\#1}$ open cavity use-case)

- ▶ **SISO unstable** model, $Re = 5250$
- ▶ $n \approx 750,000$ internal variables
- ▶ $r = 18$ internal variables
- ▶ 12 iterations $\Rightarrow$ less than 108 $x=A \backslash b$
- ▶ $\approx 1h45$ (std. laptop)

```
opt.method = 'irka'
Hr = mor.lti({A,B,C,D,E},r,opt)
```

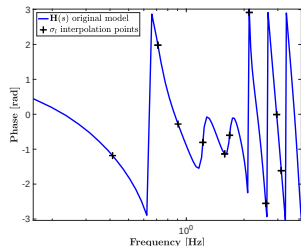
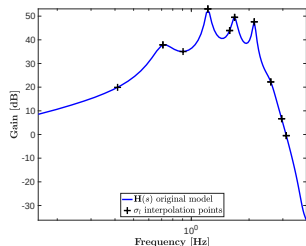


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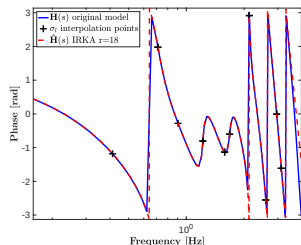
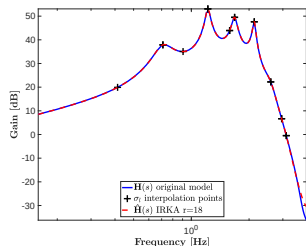
```
opt.method = 'irka'  
Hr = mor.lti({A,B,C,D,E},r,opt)
```



Dynamical model approximation and interpolation

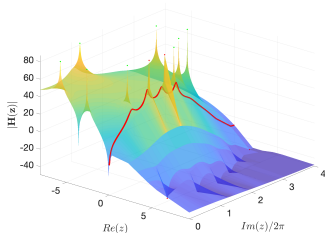
$\mathcal{H}_2$ model approximation ($H_{\#1}$ open cavity use-case)

- ▶ **SISO unstable** model, $Re = 5250$
- ▶ $n \approx 750,000$ internal variables
- ▶ $r = 18$ internal variables
- ▶ 12 iterations $\Rightarrow$ less than 108 $x=A \backslash b$
- ▶ $\approx 1h45$ (std. laptop)



```
opt.method = 'irka'
Hr = mor.lti({A,B,C,D,E},r,opt)
```

Meromorphic function at Reynolds number 5250



Dynamical model approximation and interpolation

$\mathcal{H}_2$ model approximation with I/O delay structure ($\hat{\mathbf{H}}_d := \arg \min ||\mathbf{H} - \hat{\mathbf{H}}e^{-\hat{\tau}s}||_{\mathcal{H}_2}$)

If $\hat{\mathbf{H}}$ is solution of the $\mathcal{H}_2$ approximation problem, then

$$\begin{aligned}\mathbf{H}(-\hat{\lambda}_l) &= \hat{\mathbf{H}}(-\hat{\lambda}_l) \\ \mathbf{H}'(-\hat{\lambda}_l) &= \hat{\mathbf{H}}'(-\hat{\lambda}_l)\end{aligned}$$

where $\hat{\lambda}_l := \text{eig}(\hat{E}, \hat{A})^a$ and

$$\mathbf{H}(s) = \sum_{j=1}^n \frac{\phi_j}{s - \lambda_j}$$

$$^a l = 1, \dots, r$$



I. Pontes Duff, C. P-V. and C. Seren, " *$\mathcal{H}_2$ -optimal model approximation by input / output-delay structured reduced order models*", in Systems & Control Letters, 2018, Vol. 117, pp. 60-67.

Dynamical model approximation and interpolation

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$$\mathbf{H}(s) = \sum_{j=1}^n \frac{\phi_j}{s - \lambda_j}$$

$$^a l = 1, \dots, r$$

If $\hat{\mathbf{H}}_d$ is solution of the $\mathcal{H}_2$ **input-output delayed** approximation problem, then

$$\begin{aligned}\mathbf{T}_d(-\hat{\lambda}_l) &= \hat{\mathbf{H}}(-\hat{\lambda}_l) \\ \mathbf{T}_d'(-\hat{\lambda}_l) &= \hat{\mathbf{H}}'(-\hat{\lambda}_l)\end{aligned}$$

where $\hat{\lambda}_l := \text{eig}(\hat{E}, \hat{A})$ and

$$\mathbf{T}_d(s) = \sum_{j=1}^n \frac{\phi_j}{s - \lambda_j} e^{\hat{\tau}\lambda_j}$$

$$\hat{\mathbf{H}}_d(s) = \hat{\mathbf{H}}(s)e^{-\hat{\tau}s}$$

plus a constrain on the delay $\hat{\tau}$:

$$\sum_{j=1}^n \lambda_j \phi_j \left(\sum_{k=1}^r \frac{\hat{\phi}_k}{\lambda_j + \hat{\lambda}_k} \right) e^{\hat{\tau}\lambda_j} = 0$$



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Dynamical model approximation and interpolation

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If a realisation of $\mathbf{H}$ exists, and $\hat{\tau}$ fixed

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Dynamical model approximation and interpolation

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$$\mathbf{H}(s) = \mathbf{c}(sE - A)^{-1} \mathbf{b}$$

↓ **shift**

$$\mathbf{T}_d(s) = \mathbf{c}_d(sE - A)^{-1} \mathbf{b}_d$$

plus a constrain on the delay $\hat{\tau}$:

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$$\mathbf{T}_d(s) = \mathbf{c}_d(sE - A)^{-1} \mathbf{b}_d$$

↓ **IRKA**

$$\hat{\mathbf{H}}(s) = \mathbf{c}_d V \left(sW^T E V - W^T A V \right)^{-1} W^T \mathbf{b}_d$$

plus a constrain on the delay $\hat{\tau}$:

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$$\hat{\mathbf{H}}(s) = \mathbf{c}_d V \left(sW^T E V - W^T A V \right)^{-1} W^T \mathbf{b}_d$$

↓ **optimise on $\hat{\tau}$**

$\hat{\mathbf{H}}_d = \hat{\mathbf{H}}e^{-\hat{\tau}s}$ solves the $\mathcal{H}_2$ model approx. pb.

► **IO-dITIA**

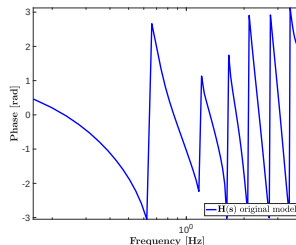
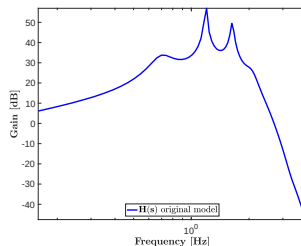


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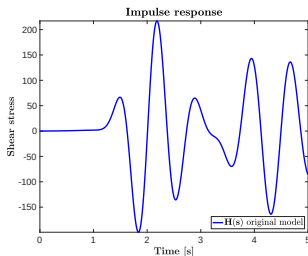
Dynamical model approximation and interpolation

$\mathcal{H}_2$ model approximation with I/O delay structure ($H_{\#1}$ open cavity use-case)

- ▶ **SISO stable** model, $Re = 4000$
- ▶ $n \approx 750,000$ internal variables
- ▶ $r = 6$ internal variables
- ▶ IRKA catches the main dynamics
- ▶ IO-dITIA also catches the delay



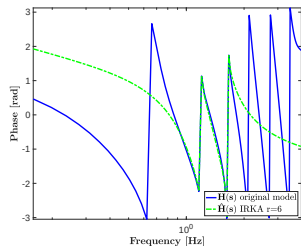
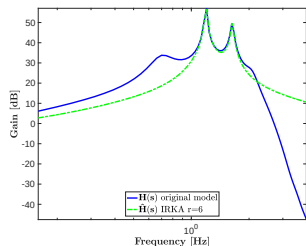
```
opt.delayIO = [0 2]  
Hr = mor.lti({A,B,C,D,E},r,opt)
```



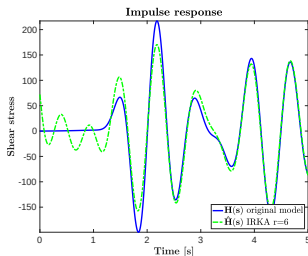
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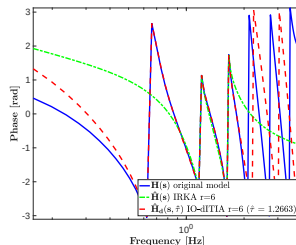
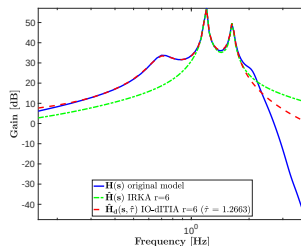
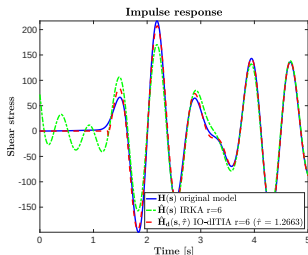


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Dynamical model approximation and interpolation

$\mathcal{H}_{2,\Omega}$ model approximation ($\hat{\mathbf{H}} := \arg \min ||\mathbf{H} - \mathbf{G}||_{\mathcal{H}_{2,\Omega}}$)

If $\hat{\mathbf{H}}$ is solution of the $\mathcal{H}_2$ approximation problem, then

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P. Vuillemin, C. P-V. and D. Alazard, "*Frequency-limited model approximation through limited $\mathcal{L}_2$ -norm minimisation*", in preparation.

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$$\begin{aligned}\mathbf{T}_\omega(-\hat{\lambda}_l) &= \hat{\mathbf{T}}_\omega(-\hat{\lambda}_l) \\ \mathbf{T}_\omega'(-\hat{\lambda}_l) &= \hat{\mathbf{T}}_\omega'(-\hat{\lambda}_l)\end{aligned}$$

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where $a_\omega(s) \triangleq \frac{2}{\pi} \mathbf{atan} \left(\frac{\omega}{s} \right)$



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- Here both $\mathbf{H}$ and $\hat{\mathbf{H}}$ are modified as $\mathbf{T}_\omega$ and $\hat{\mathbf{T}}_\omega$ (irrational)

- TF-IRKA may interpolate $\mathbf{T}_\omega$ with

$$\hat{\mathbf{H}}(s) = \hat{\mathbf{c}} (s\hat{E} - \hat{A})^{-1} \hat{\mathbf{b}}$$

- ... but one is seeking for $\hat{\mathbf{T}}_\omega$
- ... which is irrational too
⇒ need for tool (still open research)
⇒ structured reduced model



P. Vuillemin, C. P-V. and D. Alazard, "*Poles residues descent algorithm for optimal frequency-limited $\mathcal{H}_2$ model approximation*", in Proc. of the ECC, Strasbourg, France, 2014, pp. 1080-1085.



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- ▶ ... which is irrational too
⇒ need for tool (still open research)
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An alternative has been proposed

- ▶ **FL-ISTIA** ("cheap" but not optimal)
- ▶ **DARPO** (expensive, but "optimal")



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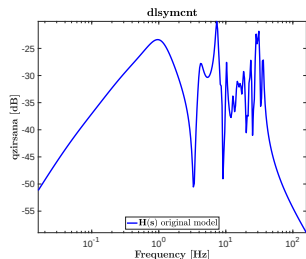
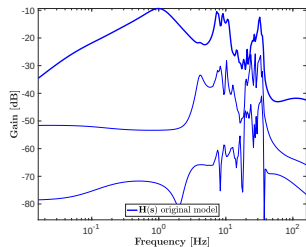


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Dynamical model approximation and interpolation

$\mathcal{H}_{2,\Omega}$ model approximation ($H_{\#2}$ vibration-oriented model use-case)

- ▶ **MIMO** aeroservoelastic model
($n \approx 700$, $n_u = 3$, $n_y = 3$)
- ▶ ... vibrations around **[7 15]Hz**
- ▶ IRKA catches the main dynamics
(but also those useless for control)
- ▶ FL-ISTIA focusses on **[0 15]Hz**

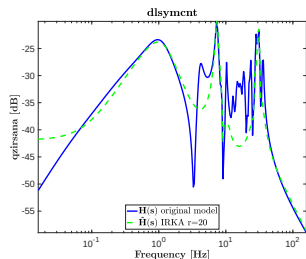
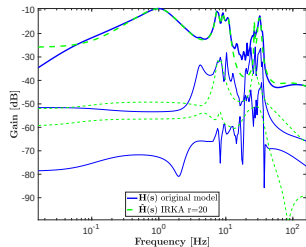


```
opt.freqBand = [0 15]*2*pi  
Hr = mor.lti(H,r,opt)
```

Dynamical model approximation and interpolation

$\mathcal{H}_{2,\Omega}$ model approximation ($H_{\#2}$ vibration-oriented model use-case)

- ▶ **MIMO** aeroservoelastic model
($n \approx 700$, $n_u = 3$, $n_y = 3$)
- ▶ ... vibrations around **[7 15]Hz**
- ▶ **IRKA** catches the main dynamics
(but also those useless for control)
- ▶ **FL-ISTIA** focusses on **[0 15]Hz**

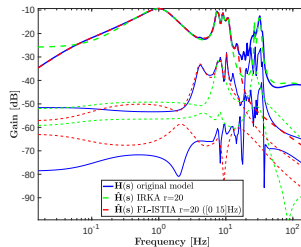


```
opt.freqBand = [0 15]*2*pi  
Hr = mor.lti(H,r,opt)
```

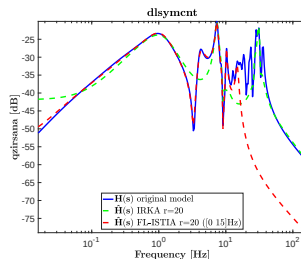
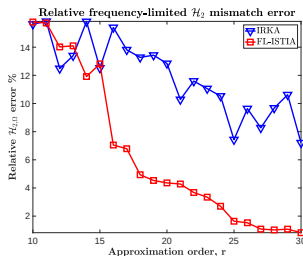
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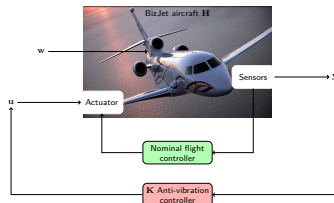
► Anti-vibration (structured) controller

$$\hat{\mathbf{H}} \xrightarrow{\text{hinfstruct}} \mathbf{K}$$

► Validated in GVT (real experiments)

$$\{\omega_i, \Phi\}_{i=1}^N \xrightarrow{\text{Loewner}} \hat{\mathbf{H}}_{\text{data}}$$

```
Hr = mor.lti({w_i, Phi_i}, 30)
```



P. Apkarian and D. Noll, "*Nonsmooth $\mathcal{H}_{\infty}$ synthesis*", IEEE Transactions on Automatic Control, 2006, Vol. 51(1), pp. 71-86.

Dynamical model approximation and interpolation

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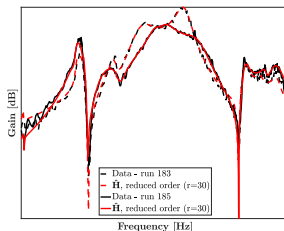
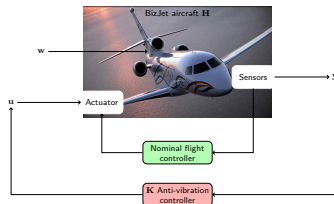
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```
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```

- ▶ Open-loop (dashed),
Closed-loop (solid)
- ▶ **GVT** $\approx 40\%$ vibration attenuation
 $\Rightarrow$ **Video (Dassault-Aviation)**
- ▶ **FT** $\approx 35\%$ vibration attenuation



A.J. Mayo and A.C. Antoulas, "*A framework for the solution of the generalized realization problem*", Linear Algebra and its Applications, 2007, Vol. 425, pp. 634-662.

Outline

Resume

Dynamical models and approximation

Dynamical model approximation and interpolation

Some open research paths

- Approximation for data-driven control

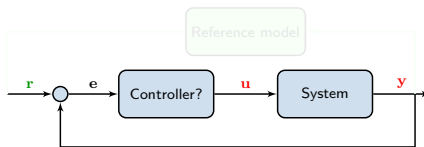
- $\mathcal{L}_2$ functions stability analysis

- Mixed continuous/sampled-time models analysis

Conclusions

Some open research paths

Approximation for data-driven control



L-DDC procedure (Loewner-Data Driven Control), involves VRFT that recasts the control synthesis problem as an identification one

- ▶ use y output identification signal to construct the fictive r reference signal
- ▶ then, $e = r - y$, and identify $e \rightarrow u$ or $u \rightarrow e$ transfer
- ▶ apply Loewner on $e = \Phi_i u$ or $u = \Phi_i^* e$ transfer

Appropriate for complex systems control (e.g. simulator...)
⇒ how addressing the stability of the closed-loop?



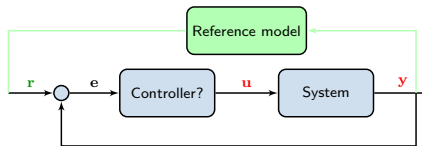
P. Kergus, C. P-V., F. Demourant and S. Formentin, "*Frequency-domain data-driven control design in the Loewner framework*", in Proc. of the IFAC WC, Toulouse, France, 2017, pp. 2095-2100.



P. Kergus, M. Olivi, C. P-V. and F. Demourant, "*From reference model selection to controller validation: application to L-DDC*", to appear in IEEE L-CSS, 2019.

Some open research paths

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Some open research paths

$\mathcal{L}_2$ functions stability analysis

How to address the stability of any $\mathbf{H} \in \mathcal{L}_2$?

Proposed statement

1. Given $\mathbf{H} \in \mathcal{L}_2$, it is possible to find $\hat{\mathbf{H}} \in \mathcal{RL}_2$ that well reproduces $\mathbf{H}$, whatever the complexity of $\mathbf{H}$ is, if we can arbitrarily increase $r = \dim(\hat{\mathbf{H}})$.
⇒ Can be obtained by increasing the Loewner matrix up to numerical rank loss.
2. If, based on an unstable realisation of $\hat{\mathbf{H}} \in \mathcal{RL}_2$, the optimal stable approximant $\hat{\mathbf{H}}_s \in \mathcal{RH}_\infty$ is close enough to $\hat{\mathbf{H}} \in \mathcal{RL}_2$, in the sense of the $\mathcal{L}_\infty$ -norm, then $\hat{\mathbf{H}}$ is stable and, following previous statement (1.), $\mathbf{H}$ is stable too.
⇒ Can be achieved by a rational stable approximation...
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$$\mathbf{H} \in \mathcal{L}_2 \xrightarrow{\text{Loewner}} \hat{\mathbf{H}} \in \mathcal{RL}_2$$



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$$\mathbf{H} \in \mathcal{L}_2 \xrightarrow{\text{Loewner}} \hat{\mathbf{H}} \in \mathcal{RL}_2 \xrightarrow[\text{proj.}]{\mathcal{RH}_\infty} \hat{\mathbf{H}}_s \in \mathcal{RH}_\infty \xrightarrow[\text{norm}]{\mathcal{L}_\infty} \|\hat{\mathbf{H}}_s - \hat{\mathbf{H}}\|_{\mathcal{L}_\infty}$$

$\downarrow$
 stable ($< \epsilon$)
 unstable (otherwise)



C. P-V. and P. Vuillemin, "On the stability estimation of $\mathcal{L}_2$ functions", in preparation.



M. Kohler, "On the closest stable descriptor system in the respective spaces $\mathcal{RH}_2$ and $\mathcal{RH}_\infty$ ", Linear Algebra and its Applications, 2014, Vol. 443, pp. 34-49.

Some open research paths

$\mathcal{L}_2$ functions stability analysis

$$H(s, p) = \frac{1}{s + e^{-ps}}$$

Theoretical stability limit $p = \pi/2$

```
H = @(s) 1./(s+exp(-p*s))
w = logspace(-2,1,100)*2*pi
stabTag = mor.stability(H,w)
```



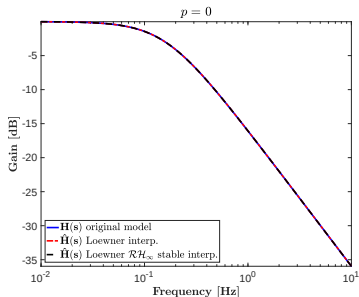
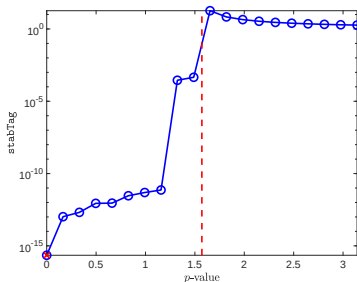
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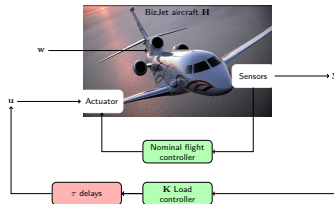


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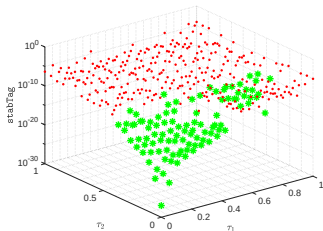
Some open research paths

$\mathcal{L}_2$ functions stability analysis ($H_{\#3}$ generic aircraft gust model use-case)

- Analyse the **delay stability margin** of the load control strategy
- **MIMO** model $n \approx 500$
- 2 gust internal delays (τ_g), plus 2 measurements delays



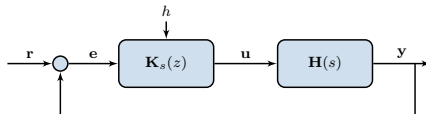
$$\mathbf{H}(s, \tau_1, \tau_2) = \mathbf{C} \left(s\mathbf{E} - \mathbf{A}_0(\tau_g) - \mathbf{A}_1(\tau_g)e^{-\tau_1 s} - \mathbf{A}_2(\tau_g)e^{-\tau_2 s} \right)^{-1} \mathbf{B}$$



```
H = @(s,tau1,tau2) = C*(s*E-...)\B
w = logspace(-1,3.5,300)
stabTag = mor.stability(H,w)
```

Some open research paths

Mixed continuous/sampled-time models analysis



$$\begin{aligned}
 \mathbf{H}(s) &= \mathbf{C} (s\mathbf{E} - \mathbf{A})^{-1} \mathbf{B} && \in \mathcal{RH}_\infty \\
 \mathbf{K}_c(\mathbf{s}) &= \mathbf{C}_c (\mathbf{s}I - \mathbf{A}_c)^{-1} \mathbf{B}_c && \in \mathcal{RH}_2 \text{ using any synth. method} \\
 \mathbf{K}_s(\mathbf{z}) &= \mathbf{C}_s (\mathbf{z}I - \mathbf{A}_s)^{-1} \mathbf{B}_s + \mathbf{D}_s && \in \mathcal{RH}_2 \text{ using any discr. method}
 \end{aligned}$$

$$s = i\omega \text{ and } z = e^{sh}, \text{ thus } s = \frac{1}{h} \ln(z)$$

Equivalently leads to

$$\begin{aligned}
 \mathbf{K}_c(\mathbf{z}) &= \mathbf{C}_c \left(\frac{1}{h} \ln(\mathbf{z})I - \mathbf{A}_c \right)^{-1} \mathbf{B}_c \\
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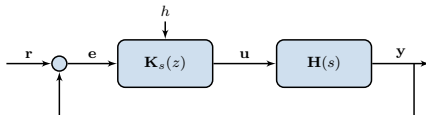
and spectrum is now repeated every π/h



C. P-V., P-L. Garoche and P. Vuillemin, "...", in preparation.

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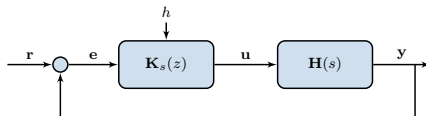
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and spectrum is now **repeated every π/h**



Some open research paths

Mixed continuous/sampled-time models analysis



Let us consider the case where $\mathbf{K}_s(z)$ is given,

$$\begin{aligned}\mathbf{K}_s(z) &= \mathbf{C}_s(z\mathbf{I} - \mathbf{A}_s)^{-1}\mathbf{B}_s \\ \mathbf{K}_s(s) &= \mathbf{C}_s(e^{sh}\mathbf{I} - \mathbf{A}_s)^{-1}\mathbf{B}_s\end{aligned}$$

Note that

$$\mathbf{u}(s) = \mathbf{K}_s(s)\mathbf{e}(s)$$

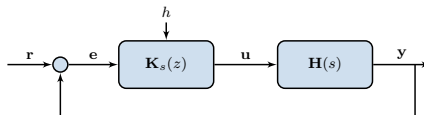
leads to a delayed algebraic

$$\mathcal{K}_s : \begin{cases} \mathbf{0} &= \mathbf{A}_s\mathbf{x}_s(t) - \mathbf{x}_s(t+h) + \mathbf{B}_s\mathbf{e}(t) \\ \mathbf{u}(t) &= \mathbf{C}_s\mathbf{x}_s(t) \end{cases}$$



Some open research paths

Mixed continuous/sampled-time models analysis



$$\begin{cases} \dot{\mathbf{x}}(t) &= A\mathbf{x}(t) + B\mathbf{u}(t) && \text{plant dyn. w. ref.} \\ \mathbf{y}(t) &= C\mathbf{x}(t) && \text{plant output} \\ 0 &= A_s\mathbf{x}_s(t) - I\mathbf{x}_s(t+h) + B_s(\mathbf{y}(t) - \mathbf{r}(t)) && \text{ctl alg. constraint w.} \\ &&& \text{sampling delay} \\ \mathbf{u}(t) &= C_s\mathbf{x}_s(t) && \text{ctl output} \end{cases}$$

$$\begin{pmatrix} E & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \dot{\mathbf{x}}(t) \\ \dot{\mathbf{x}}_s(t) \end{pmatrix} = \begin{pmatrix} A & -BC_s \\ B_sC & A_s \end{pmatrix} \begin{pmatrix} \mathbf{x}(t) \\ \mathbf{x}_s(t) \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & -I \end{pmatrix} \begin{pmatrix} \mathbf{x}(t+h) \\ \mathbf{x}_s(t+h) \end{pmatrix} + \begin{pmatrix} 0 \\ B \end{pmatrix} \mathbf{r}(t)$$

⇒ interpolation may be a tool to attack this problem?



Outline

Resume

Dynamical models and approximation

Dynamical model approximation and interpolation

Some open research paths

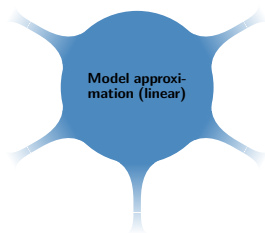
Conclusions

What to keep in mind...

... and perspectives

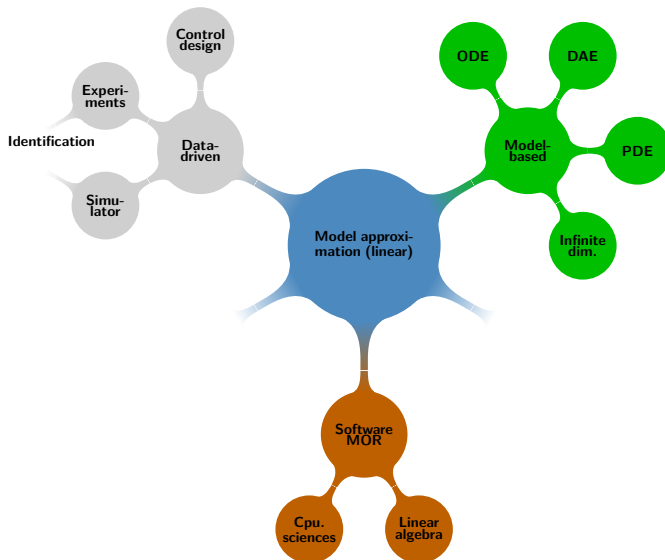
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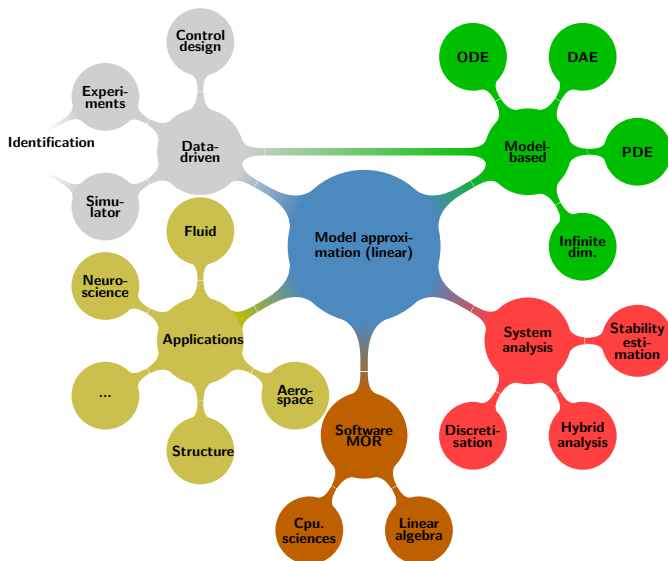
Conclusions

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Conclusions

What to keep in mind...



Conclusions

... and perspectives

Interpolation-based methods are remarkably versatile and indicated for

- ▶ model reduction and approximation,
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Integrated tool exists: MOR Toolbox at <http://mordigitalsystems.fr/>

Some interesting perspectives

- ▶ Model approximation and structures (still a lot to discover... toward nonlinear)
- ▶ Stability of $\mathcal{L}_2$ functions
- ▶ Stability of mixed continuous-sampled dynamical systems
- ▶ Discretisation issues : $\mathbf{H}(s) \leftrightarrow \mathbf{H}(z)$
- ▶ Learning methods for model construction, and control design...

→ Direct impact in engineers life and enrich their toolkit
→ May be a game changer in industrial and societal applications

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Conclusions

Acknowledgments

Ph.D. collaborators

- ▶ P. Vuillemin [approx.]
- ▶ I. Pontes Duff [approx.]
- ▶ P. Kergus [data-driven control]
- ▶ O. Kocan [ilc control]

Control mentors

- ▶ E. Mendes [M.Sc.]
- ▶ O. Sename and L. Dugard [Ph.D.]
- ▶ S.M. Savaresi [Post-Doc.]
- ▶ G. Garcia [HDR]

Industrial collaborators

- ▶ O. Cantinaud, L. Goerig [control]
- ▶ C. Meyer, G. Broux [aeroelasticity]

Academic collaborators

- ▶ D. Quero [aeroelasticity]
- ▶ A. Lepage [wt and aeroelasticity]
- ▶ D. Ossmann [observer]
- ▶ C. Carvalho, R. N. Roy [neuro-physiology]
- ▶ S. Formentin [data-driven control]
- ▶ D. Sipp, C. Leclercq [fluid mechanics]
- ▶ C. Seren, D. Alazard, A. Manecy [approximation, control and uav]

Project collaborators

- ▶ E. Coustols [projects conduction]
- ▶ M. Jungers [mosar]

Conclusions

Thanks for your attention



Large-scale dynamical model approximation and its applications

... a linear time invariant framework

Charles Poussot-Vassal

July, 12th, 2019
HDR defense



Université
de Toulouse

A.C. Antoulas	Rice University	Houston, Texas, USA	Reviewer
P. Apkarian	Onera	Toulouse, France	
G. Garcia	INSA - LAAS	Toulouse, France	
S. Grundel	Max Planck Institute	Magdeburg, Germany	
M. Olivi	INRIA Sophia-Antipolis	Nice, France	
C. Prieur	CNRS Gipsa-lab	Grenoble, France	Reviewer
K. Willcox	University of Texas	Austin, Texas, USA	Reviewer

Appendix

Interpolation-based discretisation example

- ▶ $h = 0.2$, $\omega_n \approx 31\text{rad/s}$, $\omega_N \approx 15,7\text{rad/s}$ ($f_N = 1/h/2 = 2.5\text{Hz}$)
- ▶ ‘zoh’ vs. ‘tustin’ vs. **exact** $z = e^{sh}$... $s = \ln(z)/h$

$$\mathbf{H}(s) = \frac{1}{(s^2/10 + .1s/\sqrt{10} + 1)(s^2/25 + .1s/5 + 1)}$$

$$\mathbf{H}_z(z) = \frac{0.0154z^3 + 0.1518z^2 + 0.1468z + 0.01396}{z^4 - 2.594z^3 + 3.454z^2 - 2.382z + 0.8494}$$

$$\mathbf{H}_t(z) = \frac{0.01699z^4 + 0.06798z^3 + 0.102z^2 + 0.06798z + 0.01699}{z^4 - 2.744z^3 + 3.703z^2 - 2.558z + 0.8715}$$

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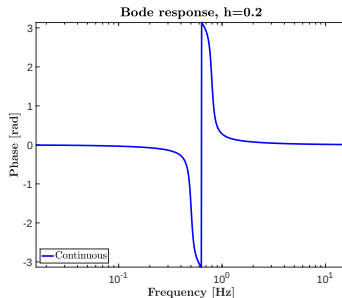
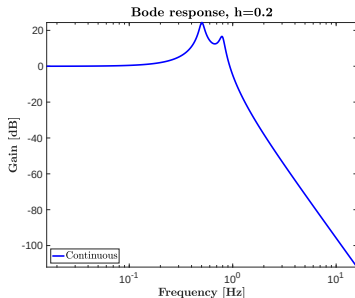
$$\mathbf{H}_e(z) = \frac{1}{\left((\ln(z)/h)^2/10 + .1\ln(z)/h/\sqrt{10} + 1\right)\left((\ln(z)/h)^2/25 + .1\ln(z)/h/5 + 1\right)}$$

⇒ **Irrational form**

Appendix

Interpolation-based discretisation example

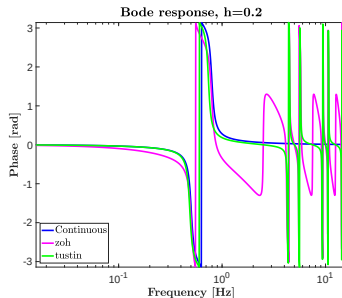
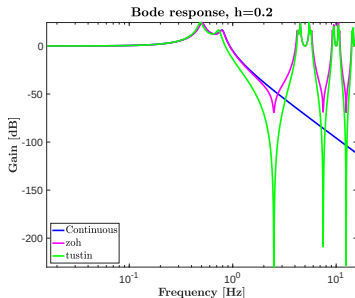
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Appendix

Interpolation-based discretisation example

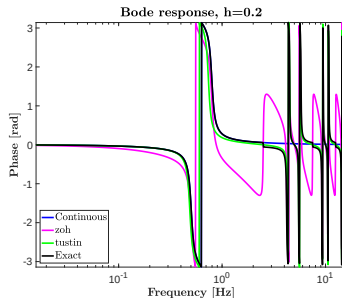
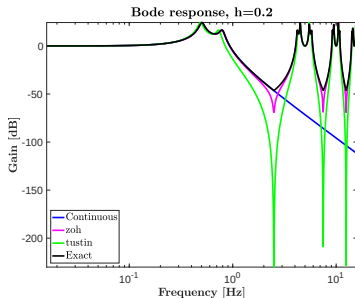
- ▶ $h = 0.2$, $\omega_n \approx 31\text{rad/s}$, $\omega_N \approx 15,7\text{rad/s}$ ($f_N = 1/h/2 = 2.5\text{Hz}$)
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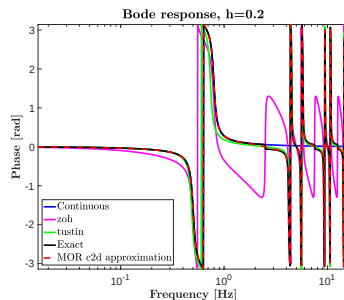
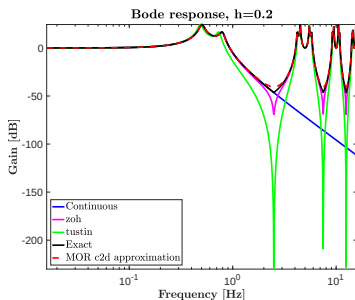
- ▶ $h = 0.2$, $\omega_n \approx 31\text{rad/s}$, $\omega_N \approx 15,7\text{rad/s}$ ($f_N = 1/h/2 = 2.5\text{Hz}$)
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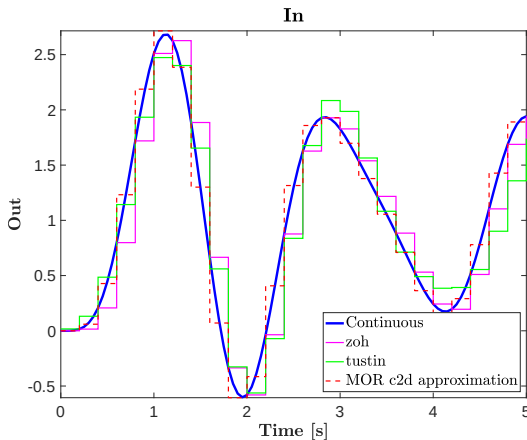
$$\mathbf{H}_r(z) = \frac{0.05964z^3 + 0.2134z^2 + 0.0552z - 0.0001461}{z^4 - 2.594z^3 + 3.454z^2 - 2.382z + 0.8494}$$

Rational approximation

Appendix

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Conclusions

Mixed continuous/sampled-time models analysis

Let be given $\mathbf{K}_s(z)$,

$$\mathbf{K}_s(z) = C_s(zI - A_s)^{-1}B_s$$

$$\mathbf{K}_s(s) = C_s(e^{sh}I - A_s)^{-1}B_s$$



Conclusions

Mixed continuous/sampled-time models analysis

Let be given $\mathbf{K}_s(z)$,

$$\begin{aligned}\mathbf{K}_s(z) &= C_s(zI - A_s)^{-1}B_s \\ \mathbf{K}_s(s) &= C_s(e^{sh}I - A_s)^{-1}B_s\end{aligned}$$

- ▶ $\mathbf{K}_s(z) \in \mathcal{RH}_2$, $\#\Lambda(A_s) = n$ and $\Lambda(A_s) \in \mathbb{D} \cup \partial\mathbb{D}$
 $\mathbf{K}_s : \mathbb{C}([0, \omega_N]) \rightarrow \mathbb{C}^{n_y \times n_u}$
- ▶ $\mathbf{K}_s(s)$, $\notin \mathcal{H}_2$, $\notin \mathcal{L}_2$ **and** $\#\Lambda(A_s, e^{sh}E_s) = \infty$
 $\mathbf{H}_s : \mathbb{C} \rightarrow \mathbb{C}^{n_y \times n_u}$



Conclusions

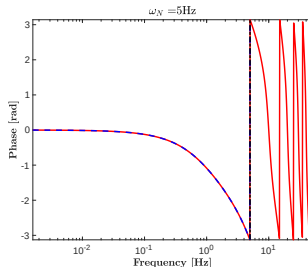
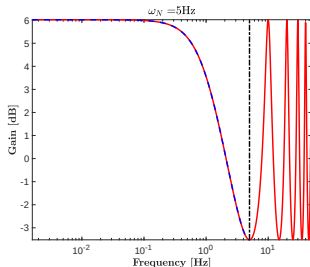
Mixed continuous/sampled-time models analysis

Let be given $\mathbf{K}_s(z)$,

$$\begin{aligned}\mathbf{K}_s(z) &= \mathbf{C}_s(z\mathbf{I} - \mathbf{A}_s)^{-1}\mathbf{B}_s \\ \mathbf{K}_s(s) &= \mathbf{C}_s(e^{sh}\mathbf{I} - \mathbf{A}_s)^{-1}\mathbf{B}_s\end{aligned}$$

$$\mathbf{K}_s(z) = \frac{1}{z - 0.5} \text{ and } \mathbf{K}_s(s) = \frac{1}{e^{sh} - 0.5}$$

```
As = .5;
h = .1;
wn = pi/h;
w = logspace(-2, log10(10*wn), 1e3);
Kd = ss(As, 1, 1, 0, h);
Ks = @(s) 1/(exp(s*h)-As);
```



C. P-V., P-L. Garoche and P. Vuillemin, "...", in preparation.